

Business Requirements Document (BRD)

Project Title: Expense Tracker with Smart Insider

1.Introduction

1.1 Purpose

This document outlines the business requirements for the development of an intelligent expense tracking system, **Expense Tracker with Smart Insider**. The purpose of this system is to provide users with not only a platform for recording expenses but also a smart solution that delivers meaningful insights, predictive analysis, and proactive financial alerts to support informed decision-making.

1.2 Project Description

The **Expense Tracker with Smart Insider** is a web-based financial management system designed to help users efficiently track, analyse, and control their daily expenses. The application goes beyond traditional expense tracking by integrating intelligent features that assist users in understanding their financial behaviour and making informed decisions.

Users can record their expenses by entering basic details such as amount, date, and description. The system leverages **Natural Language Processing (NLP)** techniques to automatically categorise these expenses into predefined categories, thereby reducing manual effort and improving accuracy.

The application provides an interactive dashboard that visually represents spending patterns using charts and summaries. These visual insights enable users to quickly identify where their money is being spent and monitor trends over time.

In addition, the system incorporates predictive analysis to estimate future expenses based on historical data. It alerts users when their current spending pattern indicates a risk of overspending. Furthermore, the system includes an anomaly detection mechanism that identifies unusual or irregular transactions and notifies users with simple explanations.

Overall, the project aims to transform expense tracking from a basic record-keeping activity into a smart, insight-driven financial management solution that enhances user awareness, control, and decision-making.

2. Business Goals

The primary business goals of the **Expense Tracker with Smart Insider** project are to develop an intelligent and user-centric financial management system that enhances the way individuals track and analyse their expenses. The system is designed to not only simplify expense recording but also provide meaningful insights that support better financial planning and control.

The key business goals are as follows:

- To provide a reliable and efficient platform for tracking daily expenses in a structured manner
- To reduce manual effort by automating expense categorisation using basic AI techniques such as NLP
- To enhance users' financial awareness by presenting data through clear and interactive visualisations
- To enable data-driven decision-making by analysing historical expense patterns
- To proactively alert users about potential overspending based on predictive analysis
- To detect unusual or abnormal financial transactions and notify users accordingly
- To improve overall user experience through a simple, intuitive, and responsive interface

3. Stakeholders

- **End Users** – Individuals who will use the system to track and manage their expenses
- **Development Team** – Responsible for system design, development, and deployment
- **Project Supervisor** – Provides guidance, evaluation, and validation of the project

No.	Stakeholder	Role
1.	End Users	Use the application
2.	Development Team	Design and implementation
3.	Project Guide	Review and validation

4. Current Challenges

- Manual tracking and categorisation of expenses is time-consuming
- Lack of predictive insights for future spending
- No alerts for overspending
- Difficulty in identifying unusual transactions

5. Proposed Solution

The proposed system aims to address the above challenges by:

- Providing a simple and user-friendly interface for recording expenses
- Automatically categorising expenses using NLP techniques
- Presenting financial data through interactive charts and dashboards
- Analysing historical data to predict future spending trends
- Generating alerts for potential overspending scenarios
- Detecting anomalies and notifying users with clear explanations

6. Scope of the Project

6.1 Included Features

- Secure user registration and login functionality
- Expense management (add, update, delete, and view records)
- Automated expense categorisation using rule-based and ML techniques
- Interactive dashboard with graphical visualisations (charts and summaries)
- Spending prediction and alert system
- Detection and reporting of unusual financial transactions

6.2 Excluded Features

- Integration with banking or financial institution APIs
- Real-time transaction synchronisation
- Payment processing or billing systems
- Mobile application development

- Advanced deep learning-based AI models
- Multi-user or shared account features

7. Business Requirements

7.1 User Requirements

- Users should be able to register and securely log into the system
- Users should be able to add, update, and delete expense records
- Users should be able to view expense summaries and insights easily
- Users should receive alerts and notifications related to spending

7.2 System Requirements

- The system should automatically categorise expenses based on input descriptions
- The system should store and manage data efficiently in a structured database
- The system should analyse historical data to generate predictions
- The system should detect unusual spending patterns and notify users
- The system should provide visual insights through charts and reports

8. Technology Stack

The system is developed using a modern full-stack architecture to ensure efficiency, scalability, and performance.

- **Frontend:** React
Used to build an interactive and user-friendly interface.
- **Backend:** FastAPI
Handles API requests, business logic, and communication between components.
- **Programming Language:** Python
Used for backend development and AI/ML implementation
- **AI/ML:** Scikit-learn
Used for expense categorisation, prediction, and anomaly detection.

- **Database:** PostgreSQL

Used for secure and structured storage of user and expense data.

No.	Layer	Technology
1.	Frontend	React
2.	Backend	FastAPI
3.	AI / ML	Python (Scikit-learn)
4.	Database	PostgreSQL

9. Process Flow (High-Level)

- User logs into the system using secure authentication
- User enters expense details (amount, description, date)
- System processes the description using NLP techniques
- Expense is automatically categorised
- Data is securely stored in the database
- Dashboard retrieves and displays updated insights
- Prediction and anomaly detection modules analyse the data
- Alerts and notifications are generated and displayed to the user

10. Assumptions

- Users will enter accurate and consistent expense data
- Users possess basic knowledge of using web applications
- A stable internet connection is available for system access
- The dataset used for AI processing is limited but adequate for basic functionality

11. Constraints

The following constraints may impact the development and performance of the system:

- The development timeline is limited to 2–3 weeks.
- The project will focus only on Minimum Viable Product (MVP) features.
- AI/ML implementation will be basic due to limited time and resources.
- The system will be developed as a web application only, not a mobile application.
- External integrations such as banking APIs will not be included.

12. Success Metrics

- High level of user satisfaction and ease of use
- Acceptable accuracy in expense categorisation
- Clear and meaningful insights from the dashboard
- Timely and relevant alerts for overspending
- Efficient system performance and responsiveness

13. Risks

- Low accuracy of machine learning models due to limited data
- Poor quality or inconsistent user input
- Limited availability of training datasets
- Performance issues under higher loads

Mitigation Strategies:

- Implement rule-based fallback mechanisms
- Apply input validation and data cleaning techniques
- Use small but well-defined datasets

- Optimise backend performance and database queries

14. Future Scope

- Development of a mobile application for wider accessibility
- Integration with banking APIs for automatic transaction tracking
- Implementation of advanced machine learning models
- Personalised financial insights and recommendations
- Voice-based expense entry and interaction